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# Simulation and Optical Design of Multilayer Transparent Conducting Oxides

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*A thesis submitted in partial fulfillment of the requirements  
for the degree of*

**Master of Science in Physics**

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2026

# Declaration

I hereby declare that the work presented in this thesis draft, entitled “Simulation and Optical Design of Multilayer Transparent Conducting Oxides” is entirely my own original work, except where due acknowledgment is made. This work has not been submitted previously, in whole or in part, to qualify for any other academic degree or award.

Date: 2<sup>nd</sup> may 26

Signature: Jayesh

# Acknowledgements

I would like to express my sincere appreciation to the Department of Physics at the Indian Institute of Technology Bombay for providing the resources and environment necessary to pursue this research. This thesis represents an independent exploration of the optical design and simulation of multilayer transparent conducting oxide structures; the process of identifying the research scope, navigating the simulation software, and executing the experimental analysis has been an immensely rewarding journey of self-directed learning.

I am grateful to Dr. Dinesh Kabra for his role as my thesis guide during this program, and I would also like to acknowledge Mr. Subir Manna for his willingness to provide mentorship.

I thank all the individuals in the lab who assisted with my work and provided insights during technical discussions. I am also deeply thankful to my friends and family for their constant support and belief in my potential throughout this challenging project. Their encouragement provided the stability I needed to see this work through to completion, and I am truly grateful for every moment of support that brought me to this milestone.

# Abstract

This thesis investigates the optical design and simulation of an Indium Zinc Oxide (IZO)/Silver (Ag)/IZO multilayer structure as a high-performance transparent conducting oxide (TCO) for solar cell applications. Traditional TCOs face a fundamental trade-off between electrical conductivity and optical transparency as their thickness decreases. By incorporating an ultrathin noble metal layer, this limitation is mitigated through the injection of charge carriers and enhanced light management via admittance matching.

Using SETFOS software, the optical constants of ultrathin silver were modified applying the Drude theory to account for Localized Surface Plasmon Resonance (LSPR) effects arising from Volmer-Weber island growth. The top and bottom IZO layers, along with the intermediate silver nanolayer, were systematically swept to optimize transmittance. For single-junction applications (300–800 nm), the optimized IZO (45 nm)/Ag (8 nm)/IZO (60 nm) structure yielded an 8.55% improvement in the theoretical short-circuit current ( $J_{sc}$ ) cap compared to a standard 140 nm IZO layer. For tandem applications (300–1300 nm), the IZO (45 nm)/Ag (8 nm)/IZO (40 nm) structure demonstrated a 13.06% transmittance enhancement.

Haacke's Figure of Merit (FoM) showed a two-to-fourfold enhancement over traditional structures. Finally, a comparative analysis of deposition techniques revealed that magnetron sputtering is significantly more material-efficient than thermal evaporation for ultrathin silver layers, requiring approximately 24 times less target mass per deposition.

# Table of Contents

|                                                                 |            |
|-----------------------------------------------------------------|------------|
| <b>Declaration</b>                                              | <b>i</b>   |
| <b>Acknowledgements</b>                                         | <b>ii</b>  |
| <b>Abstract</b>                                                 | <b>iii</b> |
| <b>List of Figures</b>                                          | <b>v</b>   |
| <b>List of Tables</b>                                           | <b>vi</b>  |
| <b>List of Abbreviations</b>                                    | <b>vii</b> |
| <b>1 Theoretical Background</b>                                 | <b>1</b>   |
| 1.1 Introduction and Literature Review . . . . .                | 1          |
| 1.2 Theory and Principles . . . . .                             | 2          |
| 1.2.1 Thin Film Metal Growth . . . . .                          | 2          |
| 1.2.2 Light Management . . . . .                                | 3          |
| 1.2.3 Admittance Matching . . . . .                             | 4          |
| <b>2 Simulation and Results</b>                                 | <b>5</b>   |
| 2.1 Simulation of Multistructure . . . . .                      | 5          |
| 2.1.1 Optical Constants and Modification . . . . .              | 5          |
| 2.1.2 Drude Theory Correction . . . . .                         | 5          |
| 2.2 Results . . . . .                                           | 6          |
| 2.2.1 Silver Transmission Variation . . . . .                   | 6          |
| 2.2.2 IZO/Ag/IZO Simulation . . . . .                           | 7          |
| 2.3 Analysis of Results . . . . .                               | 9          |
| 2.3.1 Short-Circuit Current Calculations ( $J_{sc}$ ) . . . . . | 9          |
| 2.3.2 Resistance of the Multilayer Structure . . . . .          | 9          |
| 2.3.3 Figure of Merit (FoM) . . . . .                           | 10         |
| <b>3 Deposition Method Analysis</b>                             | <b>12</b>  |
| 3.1 Motivation . . . . .                                        | 12         |
| 3.2 Cost Approximation of Silver Deposition . . . . .           | 13         |
| 3.2.1 Evaporation Tool . . . . .                                | 13         |
| 3.2.2 Sputtering Tool . . . . .                                 | 13         |
| 3.3 Conclusion on Deposition Methods . . . . .                  | 14         |
| <b>4 Conclusion and Future Outlook</b>                          | <b>15</b>  |
| 4.1 Identified Research Gaps . . . . .                          | 15         |
| 4.2 Conclusion . . . . .                                        | 15         |
| <b>References</b>                                               | <b>16</b>  |

# List of Figures

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1.1 | Visual representations of Volmer-Weber island formation and uniform layer growth. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2  |
| 1.2 | SEM comparisons of non-wetting vs. Al 1nm wetting 8nm silver layers. Deposition on k9 glass (image ref. 14) . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3  |
| 1.3 | a monolayer and intensity transmission . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3  |
| 1.4 | Reflection vectors and interface geometry (Adapted from Ref. [4]). . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4  |
| 2.1 | <b>Simulated Transmittance of Silver Nanolayers:</b> Optical transmittance at 500 nm for silver films of varying thicknesses (2–16 nm) deposited on a glass substrate. A 2 nm Germanium (Ge) wetting layer was modeled for the 2–8 nm range to ensure uniformity, while bulk silver optical constants were utilized for 8–16 nm. The sharp drop in transmittance at 8 nm reflects the structural transition from isolated island formation (Volmer-Weber growth) to a continuous film, where absorption rapidly increases due to Localized Surface Plasmon Resonance (LSPR). . . . . | 6  |
| 2.2 | <b>Top IZO Layer Thickness Optimization (300–1300 nm):</b> Simulated transmittance spectra optimizing the top IZO layer in an IZO/Ag/IZO stack. The intermediate silver layer and bottom IZO layer are fixed at 8 nm and 40 nm, respectively, while the top IZO layer is swept from 20 nm to 60 nm. Panel (a) displays the full broadband spectra, and (b) provides a focused view of the peak transmittance region, identifying 45 nm as the optimal top layer thickness for maximum light transmission. . . . .                                                                    | 7  |
| 2.3 | <b>Silver Nanolayer Thickness Optimization:</b> Simulated transmittance spectra evaluating the intermediate silver layer thickness (swept from 6 nm to 16 nm). The top IZO layer is fixed at the previously optimized 45 nm, and the bottom IZO is fixed at 40 nm. The results highlight the optical attenuation tradeoff as silver thickness increases beyond 8 nm. . . . .                                                                                                                                                                                                         | 8  |
| 2.4 | <b>Optical Performance Comparison of Optimized Multilayers vs. Traditional TCOs:</b> (a) Broadband evaluation comparing the optimal tandem-cell structure [IZO (45 nm)/Ag (8 nm)/IZO (40 nm)] against a standard 140 nm IZO layer. (b) Visible spectrum evaluation comparing the optimal single-junction structure [IZO (45 nm)/Ag (8 nm)/IZO (60 nm)] against both standard 140 nm IZO and ITO layers. The multilayer structures demonstrate superior peak transmittance over traditional layers due to effective admittance matching. . . . .                                      | 8  |
| 2.5 | a. Schematic of lateral resistance measurement on the IZO/Ag/IZO thin film. And b is a multilayer structure . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10 |
| 3.1 | Material loss observed in the evaporation chamber (number of depositions-unknown). . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 12 |

# List of Tables

|     |                                                                       |    |
|-----|-----------------------------------------------------------------------|----|
| 2.1 | Performance comparison over the 300–800 nm wavelength range. . . . .  | 9  |
| 2.2 | Performance comparison over the 300–1300 nm wavelength range. . . . . | 9  |
| 2.3 | Haacke’s Figure of Merit over the 300–800 nm range. . . . .           | 11 |
| 2.4 | Haacke’s Figure of Merit over the 300–1300 nm range. . . . .          | 11 |

# List of Abbreviations

**DMD** Dielectric-Metal-Dielectric

**FoM** Figure of Merit

**IQE** Internal Quantum Efficiency

**ITO** Indium Tin Oxide

**IZO** Indium Zinc Oxide

**LSPR** Localized Surface Plasmon Resonance

**NCPRE** National Centre for Photovoltaic Research and Education

**OMO** Oxide-Metal-Oxide

**PVD** Physical Vapor Deposition

**TCO** Transparent Conducting Oxide



# Chapter 1

## Theoretical Background

### 1.1 Introduction and Literature Review

Traditionally used Indium Zinc Oxide (IZO) has limitations as its thickness decreases and its sheet resistance starts to increase. However, the thinner the nanolayer, the more transparent it becomes. This creates a bottleneck regarding the minimum thickness of the Transparent Conducting Oxide (TCO). Traditionally, the optimal thickness of a TCO used in solar cell applications is around 150 nm. The reason for the resistance increase in thinner layers is a consequence of the carrier decrease.

How can we reduce the thickness to achieve greater transparency without sacrificing conductivity? By compensating for the reduced carrier density, the primary solution that comes to mind is to inject charge carriers into the structure. Metals such as silver (Ag), gold (Au), copper (Cu), and aluminum (Al) are excellent candidates for this. Another viable solution is to use multilayers of highly efficient TCOs, such as alternating layers of IZO and ITO. While the latter option does not provide extra charge carriers, light management—specifically admittance matching—has demonstrated improved transparency in the visible range.

Bae *et al.* [1] studied the IZO/Ag/IZO multistrukture, where all layers were deposited using sputtering. They discussed the disadvantages of a wetting layer and demonstrated that flash-lamp annealing treatment can suppress island growth, experimentally showing a 4% improvement in short-circuit current ( $J_{sc}$ ) in comparison with a solar cell with an FTO-based TCO, without the use of a wetting layer. Lee *et al.* [2] studied an IZO/ITO double-layered structure; the ITO thickness was fixed, and then an IZO layer was deposited on it. As the IZO thickness varied, it shifted the reflection peak further into the optical range, and an improvement in  $J_{sc}$  was observed.

Zhang *et al.* [3], in a rigorous review, discussed the optical design of thin metal film-based TCOs. Dielectric-Metal-Dielectric (DMD) and DDMD structures, along with the theory for reflection simulation, were thoroughly discussed. Ji *et al.* [4] detailed the mathematics for anti-reflection theory, formulating the reflection coefficient for DMD structures. They constructed an  $\text{Al}_2\text{O}_3/\text{Ag}(\text{Cu-doped})/\text{ZnO}$  multistrukture and showed improvements in both visibility and flexibility. In their study, a 6 nm Ag layer was sputter-deposited. Doping the Ag with Cu minimized island growth even at a 6 nm thickness. Multiple studies show consistent improvements in TCOs through the use of multilayer structures [5, 6, 7, 8, 9, 10].

From the literature, we can conclude that a highly promising multilayer structure is IZO/Metal/IZO. A recent and popular replacement for traditional ITO is IZO. The best choice of metal here would be silver due to its exceptional conductivity and low absorption. Other metals could be explored further for cost efficiency.

## 1.2 Theory and Principles

### 1.2.1 Thin Film Metal Growth

Metal growth usually follows a few distinct steps: first, nucleation occurs, followed by coalescence (the physical process where two or more droplets, bubbles, or particles merge upon contact to form a single, larger entity), and finally, thickness growth begins [8].

Materials possess inherent surface energy. Three parameters are defined to categorize the strength of surface energy:  $\gamma_s$  (the surface energy of the substrate),  $\gamma_m$  (the surface energy of the metal), and  $\gamma_{m/s}$  (the metal-substrate interface energy). Depending on which surface energy dominates, there are three types of growth modes [11]. If the substrate has a higher surface energy relative to the other two terms, atoms deposit uniformly on the surface rather than nucleating, resulting in layer-by-layer growth. Conversely, if the metal surface energy ( $\gamma_m$ ) is dominant, metal atoms stick to each other, forming an island-like structure. If the interface energy is high, initial uniform layers are formed, followed by subsequent island growth [11].

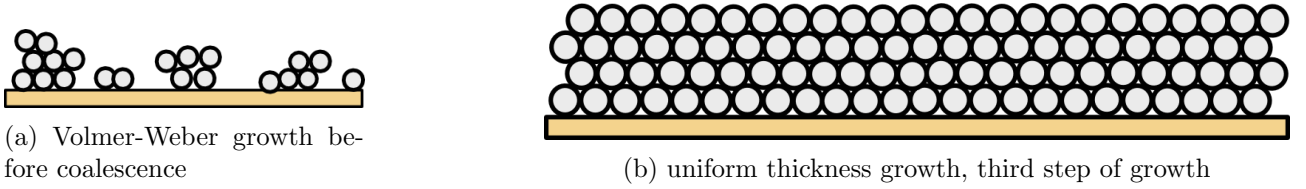


Figure 1.1: Visual representations of Volmer-Weber island formation and uniform layer growth.

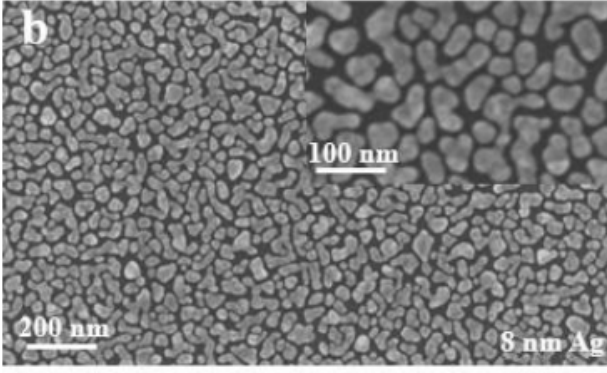
The adhesion energy ( $E_{adh}$ ) is defined as:

$$E_{adh} = \gamma_m + \gamma_s - \gamma_{m/s} \quad (1.1)$$

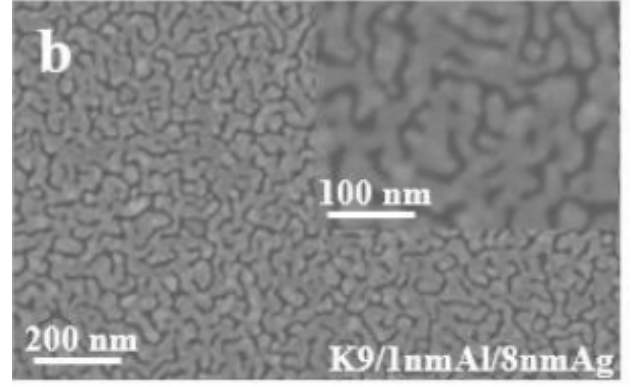
Here,  $E_{adh}$  is the energy required to separate the metal-substrate interface in a vacuum [12]. Alternatively, it can be viewed as the energy keeping the metal and substrate together. When  $\gamma_m$  becomes greater than the adhesion energy, the growth follows the Volmer-Weber method (island formation). Metals like Ag, Au, and Cu have a higher metal surface energy than adhesion energy (when the substrate is  $\text{SiO}_2$ ), whereas Al has a higher adhesion energy.

Island formation creates non-uniformity in nanolayers. Studies on silver nanolayers conclude that after reaching 8-10 nm in thickness, the silver layer starts becoming uniform (coalescing); however, increasing the thickness further starts to decrease transparency [13].

To achieve a more uniform structure, a wetting layer is deposited. Wetting layers provide additional adhesion energy because surface atoms can bond on one side via their dangling bonds, thereby satisfying the requirement for stable sites. When metal deposition begins, these stable sites serve as nucleation sites, significantly reducing island formation. Li *et al.* [14], in a concise literature survey, mentions different studies on wetting layer materials along with other methods to increase the uniformity of the silver metal layer. The most commonly used materials for wetting layers are Cr, Ti, Cu, and Ni. Typically, the wetting layer thickness is restricted to 1-2 nm, as it does not provide any functionality other than extra surface nucleation [14].



(a) 8nm Ag, without wetting



(b) 8nm Ag, with 1nm Al wetting

Figure 1.2: SEM comparisons of non-wetting vs. Al 1nm wetting 8nm silver layers. Deposition on k9 glass (image ref. 14)

### 1.2.2 Light Management

When we evaluate a thin layer, we look at its most important optical properties: reflection ( $R$ ), absorption ( $A$ ), and transmittance ( $T$ ). Transmittance is given as:

$$T = 1 - R - A \quad (1.2)$$

Absorption within the layer is calculated as [3]:

$$A = \frac{4\pi nk}{\lambda_0} \int_0^d (|E(z)|)^2 dz \quad (1.3)$$

Where the complex permittivity of the metal is defined as:

$$\epsilon_{metal} = \epsilon' + i\epsilon'', \quad \text{where} \quad \begin{cases} \epsilon' = n^2 - k^2 \\ \epsilon'' = 2nk \end{cases} \quad (1.4)$$

Here, the electric field  $E(z)$  is set up between layers,  $d$  is the thickness of the layer, and  $\epsilon$  is the permittivity. The complex refractive index is denoted as  $\tilde{n} = n + ik$ , where  $k$  is the extinction coefficient, which dictates how strongly the material attenuates the incoming light intensity. Looking at the above equation, there are multiple strategies to reduce absorption in a material, such as selecting a material with a low  $nk$  product or utilizing a thinner layer. By examining the permittivity graphs of various materials, it is evident that silver has the lowest  $nk$  product compared to Cu, Al, and Au.

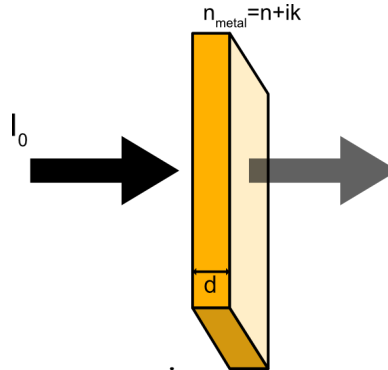


Figure 1.3: a monolayer and intensity transmission .

### 1.2.3 Admittance Matching

We now understand the physics of wetting layers and metal growth. The next step is determining how to improve the transparency of the TCO by reducing its thickness while compensating for charge carriers with silver. If we look at the transparency variation of metal layers like silver, we see a massive reflection loss compared to absorption loss in the visible range. These are the losses to be suppressed.

When designing multistuctures, we introduce multiple interfaces, which increases the interface interaction of light (e.g., reflection). We aim to reduce this back-reflection. This is achieved when reflected light from multiple layers destructively interferes at the boundary to maximize admittance ( $Y = H/E$ ). When back-reflected light does not leave the boundary of the layer, the forward propagation of incident light increases. This process is called admittance matching.

Instead of an experimental trial-and-error approach involving multiple deposition cycles, the transfer matrix method is used to find the optimal thickness combination for the IZO/Ag/IZO structure. Ji *et al.* [4] discussed the formulation for reflection minimization in thin-layer multistuctures. The reflection coefficient between two layers is given by:

$$r_{pq} = \frac{\tilde{n}_p \cos \theta_p - \tilde{n}_q \cos \theta_q}{\tilde{n}_p \cos \theta_p + \tilde{n}_q \cos \theta_q} \quad (1.5)$$

Where  $\tilde{n}$  is the complex refractive index of the material, and  $\theta$  is the angle relative to the surface normal.

Because silver has a low  $nk$  product due to a negligible real part of the refractive index,  $\tilde{n}$  becomes purely imaginary in the metal layer. This results in exponential attenuation rather than oscillation. The overall reflection phase vector  $r_{3,45}$  represents attenuation due to the third layer and the phase shift from the second dielectric:

$$r_{3,45} = \frac{r_{34} + r_{45} \exp\left(i \frac{4\pi}{\lambda} \tilde{n}_4 d_4\right)}{1 + r_{34} r_{45} \exp\left(i \frac{4\pi}{\lambda} \tilde{n}_4 d_4\right)} \quad (1.6)$$

As penetration increases, absorption also increases because light must pass through multiple media. The necessary condition for reflection minimization is to ensure that the magnitude of the  $r_{3,45}$  vector is greater than the combined magnitude of the other two destructively interfering waves, allowing them to extinguish themselves:

$$\left| r_{3,45} \exp\left(-\frac{4\pi}{\lambda} k_3 d_3\right) \exp\left(i \frac{4\pi}{\lambda} \tilde{n}_2 d_2\right) \right| \geq \left| r_{23} \exp\left(i \frac{4\pi}{\lambda} \tilde{n}_2 d_2\right) \right| - |r_{12}| \quad (1.7)$$

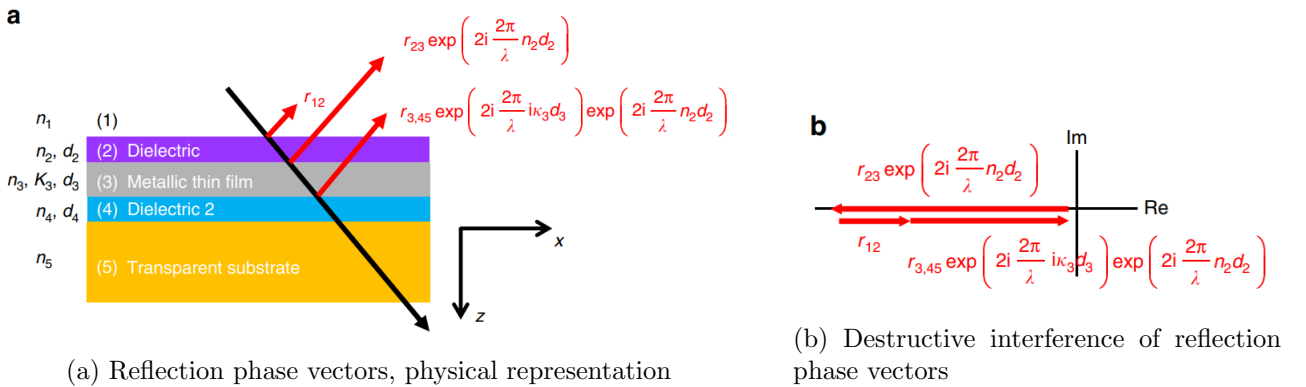


Figure 1.4: Reflection vectors and interface geometry (Adapted from Ref. [4]).

## Chapter 2

# Simulation and Results

### 2.1 Simulation of Multistructure

SETFOS (Semiconducting Thin Film Optics Simulation) is an advanced software platform used for sophisticated semiconductor simulations. For our research, the optical absorption module was utilized. In this module, the forward and backward traveling electric fields are calculated using the transfer matrix method.

#### 2.1.1 Optical Constants and Modification

To simulate a material's optical properties in SETFOS, the complex refractive index ( $\tilde{n} = n + ik$ ) must be input. As discussed,  $n$  and  $k$  account for light transmission as well as loss within the media. For this work, IZO ellipsometry data were extracted from a sputter-deposited, optimized lab sample of 140 nm. The ITO  $n$  and  $k$  data were taken from standard files provided within the software.

However, obtaining accurate data for the silver nanolayer is challenging. The  $n$  and  $k$  values vary considerably between bulk silver and nanolayer silver. Below 20 nm, the  $n$  and  $k$  data deviate drastically from bulk values due to island formation. As surface roughness increases below 10 nm, the islands absorb light at specific frequencies due to Localized Surface Plasmon Resonance (LSPR) [3].

While performing optical ellipsometry on a lab-created nanolayer is the best option, nanolayer silver is highly reactive due to its large available contact area. Furthermore, the availability of  $n$  and  $k$  data for silver layers below 10 nm (particularly with a wetting layer) is rare in the literature. Therefore, here I used theoretical models to modify the  $n$  and  $k$  data and verified the results against the available data.

#### 2.1.2 Drude Theory Correction

Base data for a 20 nm silver layer deposited over a wetting layer was used as a reference, with  $n$  and  $k$  obtained from the provided complex permittivity data [15]. To obtain accurate  $n$  and  $k$  data at 10 nm, corrections were applied using the Drude theory for thin films:

$$\epsilon = \epsilon' + i\epsilon'', \quad \tilde{n}^2 = (n + ik)^2 \quad (2.1)$$

$$n = \sqrt{\frac{\sqrt{\epsilon'^2 + \epsilon''^2} + \epsilon'}{2}}, \quad k = \sqrt{\frac{\sqrt{\epsilon'^2 + \epsilon''^2} - \epsilon'}{2}} \quad (2.2)$$

The Drude model describes the permittivity as:

$$\epsilon(\omega) = \epsilon_\infty - \frac{\omega_p^2}{\omega^2 + i\gamma\omega} \quad (2.3)$$

Where  $\omega_p$  is the plasma frequency,  $\gamma$  is the electron damping rate, and  $\epsilon_\infty$  is the high-frequency bound-electron contribution. From this, we understand that  $\epsilon''(\omega) \propto \gamma$ .

As the thickness decreases from 20 nm to 10 nm, the damping rate  $\gamma$  increases. This thickness-dependent damping is given by:

$$\gamma(t) = \gamma_0 + \frac{Av_F}{t} \quad (2.4)$$

$Av_F$  is effective electron velocity factor.

We can approximate the modified imaginary permittivity at 10 nm as:

$$\epsilon''_{10\text{nm}} = F \cdot \epsilon''_{20\text{nm}} \quad (2.5)$$

Where the scaling factor  $F$  is approximated by:

$$F = \frac{\gamma(10)}{\gamma(20)} = \frac{\gamma_0 + \frac{Av_F}{10}}{\gamma_0 + \frac{Av_F}{20}} \quad (2.6)$$

This factor  $F$  is calculated to be approximately 1.5, from which the updated  $n$  and  $k$  values for the 10 nm layer are derived. Note that the value of  $F$  is an estimate, made by assuming that for a 20 nm thickness, the term  $Av_F/t$  is roughly equal to  $\gamma_0$ . In other words, the correction factor for a 20 nm layer is approximately equal to the bulk damping factor.

We utilized this well-established Drude theory correction for thin films following the methodologies outlined in the study of optical constants of silver by Jiang *et al.* [16].

## 2.2 Results

### 2.2.1 Silver Transmission Variation

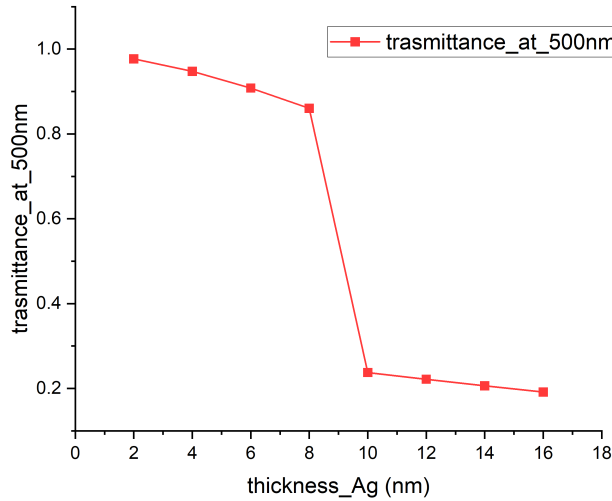


Figure 2.1: **Simulated Transmittance of Silver Nanolayers:** Optical transmittance at 500 nm for silver films of varying thicknesses (2–16 nm) deposited on a glass substrate. A 2 nm Germanium (Ge) wetting layer was modeled for the 2–8 nm range to ensure uniformity, while bulk silver optical constants were utilized for 8–16 nm. The sharp drop in transmittance at 8 nm reflects the structural transition from isolated island formation (Volmer-Weber growth) to a continuous film, where absorption rapidly increases due to Localized Surface Plasmon Resonance (LSPR).

The transmission graph was plotted using  $n$  and  $k$  data for silver (with a 2 nm Ge wetting layer) deposited on a glass substrate for 2–8 nm [15]. Bulk silver  $n$  and  $k$  data were used for thicknesses of 8–16 nm. A prominent kink can be seen at 8 nm. The reason for this is attributed to the discontinuous

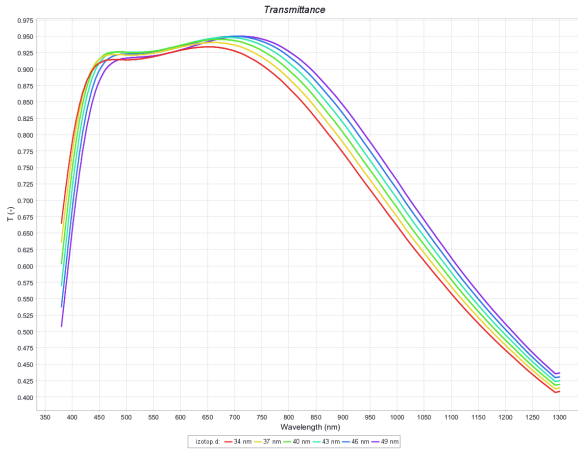
structure of the silver due to Volmer-Weber growth. The formed island structures absorb light at a particular wavelength due to Localized Surface Plasmon Resonance (LSPR), which gives us variation in the optical constants.

Due to its extremely small thickness (1–2 nm), the added wetting layer does not contribute to direct conductivity or optical impedance, but the increased uniformity of the silver layer does contribute to the overall output. Hence, all the simulation and analysis were performed using the  $n$  and  $k$  data of the complete structure: a 10 nm silver layer plus a 2 nm Ge wetting layer. Calculations for the 10 nm silver layer are provided in the preceding simulation section.

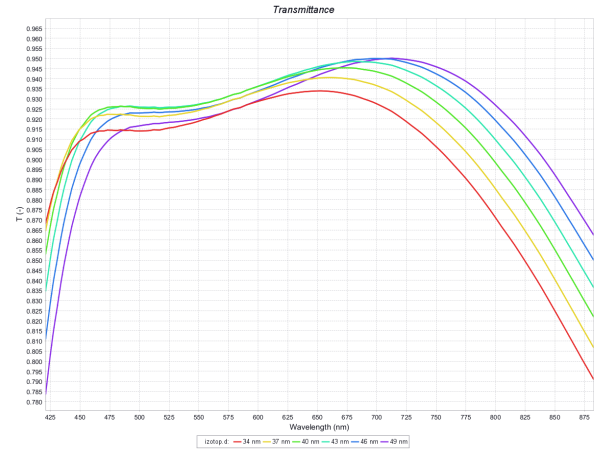
### 2.2.2 IZO/Ag/IZO Simulation

The IZO/Ag/IZO multistructure can be used to transmit only the visible spectrum as well as the combined visible and IR spectrum, according to the type of solar cell. Hence, the optimization of thickness and all related calculations were done for both regimes.

The sweep module was primarily used for the simulation. First, both the top and bottom layers were swept for 20–60 nm thicknesses while keeping the silver layer constant. Then, the base IZO layer was kept at 40 nm for the best results, and the top IZO layer was varied from 20–60 nm in thickness.



(a) T vs wavelength



(b) zoomed in

**Figure 2.2: Top IZO Layer Thickness Optimization (300–1300 nm):** Simulated transmittance spectra optimizing the top IZO layer in an IZO/Ag/IZO stack. The intermediate silver layer and bottom IZO layer are fixed at 8 nm and 40 nm, respectively, while the top IZO layer is swept from 20 nm to 60 nm. Panel (a) displays the full broadband spectra, and (b) provides a focused view of the peak transmittance region, identifying 45 nm as the optimal top layer thickness for maximum light transmission.

As we can see, the best top IZO thickness is around 45 nm. Because the sunlight intensity distribution is not constant, the developed method for the  $J_{sc}$  upper cap was used for thickness selection near the 45 nm thickness value.

Following this, the silver thickness was varied from 6–16 nm.

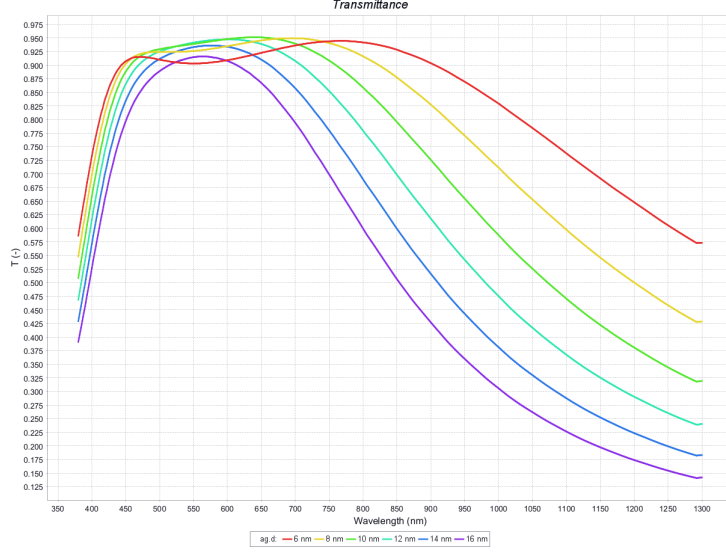
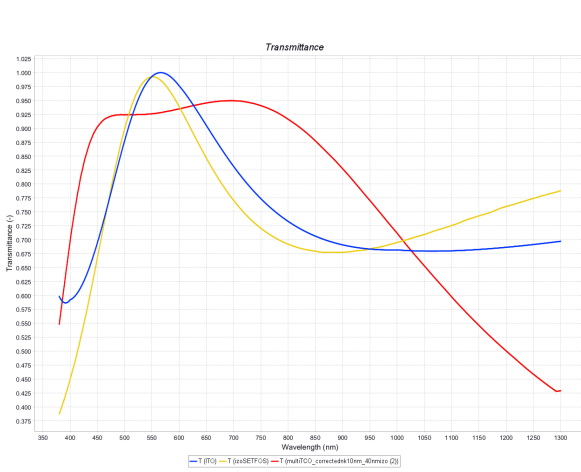
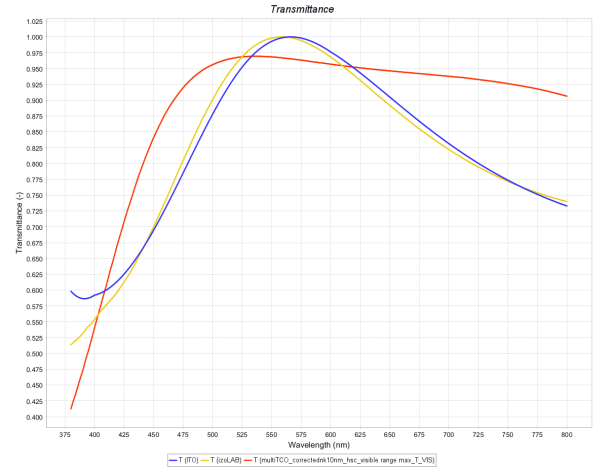


Figure 2.3: **Silver Nanolayer Thickness Optimization:** Simulated transmittance spectra evaluating the intermediate silver layer thickness (swept from 6 nm to 16 nm). The top IZO layer is fixed at the previously optimized 45 nm, and the bottom IZO is fixed at 40 nm. The results highlight the optical attenuation tradeoff as silver thickness increases beyond 8 nm.

Similarly, for the visible range only, the simulation was performed between 300–800 nm wavelengths. The optimized thickness combination was noted as IZO (45 nm)/Ag (8 nm)/IZO (60 nm). Below graphs show the comparison with traditional TCO, improved admittance is clearly visible in the overlaying graphs.



(a) TCO 300-1300nm wavelength



(b) TCO 300-800nm wavelength

Figure 2.4: **Optical Performance Comparison of Optimized Multilayers vs. Traditional TCOs:** (a) Broadband evaluation comparing the optimal tandem-cell structure [IZO (45 nm)/Ag (8 nm)/IZO (40 nm)] against a standard 140 nm IZO layer. (b) Visible spectrum evaluation comparing the optimal single-junction structure [IZO (45 nm)/Ag (8 nm)/IZO (60 nm)] against both standard 140 nm IZO and ITO layers. The multilayer structures demonstrate superior peak transmittance over traditional layers due to effective admittance matching.

*Note:* To remove clutter in the simulation graphs due to imperfect ITO variation at the 300 nm



wavelength, the simulation output is kept from 380 nm onwards.

## 2.3 Analysis of Results

### 2.3.1 Short-Circuit Current Calculations ( $J_{sc}$ )

Having observed an improvement in transparency, it is essential to quantify the resulting enhancement in the actual short-circuit current ( $J_{sc}$ ). However, the SETFOS software cannot directly simulate a multilayer TCO stack. A fundamental rule for this simulation module is that semiconducting layers must exist between two electrodes, and we cannot define the entire IZO/Ag/IZO structure simply as an electrode.

Therefore, I developed an analytical method that bypasses the need for an additional complex device simulation and directly yields the  $J_{sc}$ . We treat the IZO/Ag/IZO stack as an optical filter that attenuates a fraction of the incoming photons. Using the simulated transmittance data—which is the ratio of output to input intensity and directly proportional to the photon count—we calculate the total number of transmitted photons. Assuming an internal quantum efficiency (IQE) of 100% (where every transmitted photon generates one electron), the number of photons  $N$  is calculated as:

$$N = \int T(\lambda) \frac{I(\lambda)\lambda}{hc} d\lambda \quad (2.7)$$

Where  $N$  is the number of photons transmitted per unit area per unit time. Multiplying this by the elementary charge of an electron yields the  $J_{sc}$ . This value represents the upper theoretical cap for  $J_{sc}$  due to the assumption of a perfect IQE.

Weighted average transmission calculated as ,

$$\langle T \rangle = \frac{\int T(\lambda) I(\lambda) d\lambda}{\int I(\lambda) d\lambda} \quad (2.8)$$

Two wavelength ranges were examined according to their respective applications. For a single-junction solar cell, only the visible range is necessary; for a tandem cell, both visible and infrared (IR) light are required.

Table 2.1: Performance comparison over the 300–800 nm wavelength range.

| Structure                         | Weighted Average Transmittance (%) | $J_{sc}$ Cap (mA/cm <sup>2</sup> ) | Photons / (area · time) |
|-----------------------------------|------------------------------------|------------------------------------|-------------------------|
| IZO (140 nm)                      | 78.7                               | 42.1                               | $1.71 \times 10^{21}$   |
| IZO (45 nm)/Ag (8 nm)/IZO (60 nm) | 85.2                               | 45.7                               | $1.855 \times 10^{21}$  |
| <b>Improvement</b>                | <b>8.26%</b>                       | <b>8.55%</b>                       | <b>8.48%</b>            |

Table 2.2: Performance comparison over the 300–1300 nm wavelength range.

| Structure                         | Weighted Average Transmittance (%) | $J_{sc}$ Cap (mA/cm <sup>2</sup> ) | Photons / (area · time) |
|-----------------------------------|------------------------------------|------------------------------------|-------------------------|
| IZO (140 nm)                      | 71.2                               | 71.2                               | $2.89 \times 10^{21}$   |
| IZO (45 nm)/Ag (8 nm)/IZO (40 nm) | 80.5                               | 75.6                               | $3.07 \times 10^{21}$   |
| <b>Improvement</b>                | <b>13.06%</b>                      | <b>6.17%</b>                       | <b>6.23%</b>            |

Consequently, achieved an approximate 8% improvement in  $J_{sc}$  within the visible range, and a 13% improvement in transmittance (leading to a  $> 6\%$   $J_{sc}$  increase) over the extended visible and IR range.

### 2.3.2 Resistance of the Multilayer Structure

In laboratory settings, a four-probe setup is typically used for sheet resistance measurements. The probes are placed horizontally on the thin film surface (see Figure 2.5). However, in practical device

applications, current often flows perpendicular to the thin film plane. Why, then, do we rely on lateral measurements?

Because thin films have a minuscule thickness, vertical resistance is often negligible as there is minimal material to impede electron flow. Furthermore, when top electrode fingers are utilized in a solar cell, the majority of charge carriers must travel laterally along the film to reach the contacts, making lateral conductivity highly significant.

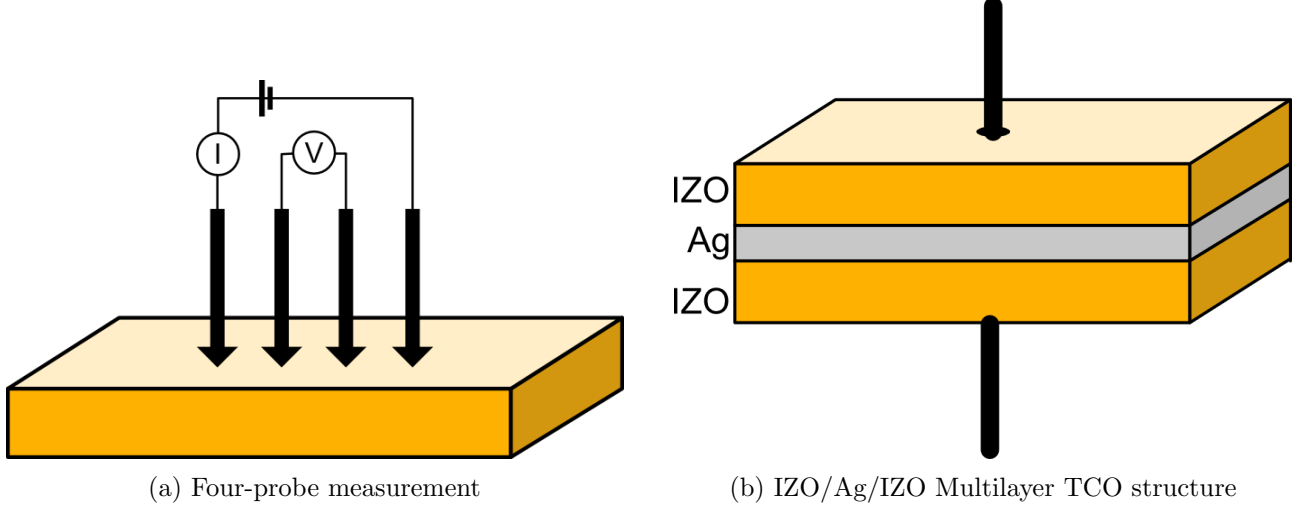


Figure 2.5: a. Schematic of lateral resistance measurement on the IZO/Ag/IZO thin film. And b is a multilayer structure

For the thin-film multilayer TCO, one might intuitively model the layers as a series of resistors. However, because we are measuring and utilizing lateral resistivity, the current passes through the layers horizontally. Thus, we employ the parallel resistance mixture rule to calculate the overall sheet resistance:

$$\frac{1}{R_{sh}} = \frac{t_1}{\rho_{IZO,1}} + \frac{t_{Ag}}{\rho_{Ag}} + \frac{t_2}{\rho_{IZO,2}} \quad (2.9)$$

For the 300-1300 nm wavelength range, the sheet resistance of a 10 nm continuous silver film, calculated with a 2 nm germanium wetting layer, is reported to be  $22 \, \Omega/\square$  [22]. This corresponds to a resistivity ( $\rho_{Ag}$ ) of  $5.1 \times 10^{-7} \, \Omega \cdot m$ . Our optimized structure consists of an 8 nm Ag layer sandwiched between a 45 nm top IZO layer and a 40 nm bottom IZO layer. The resistivity of IZO ( $\rho_{IZO}$ ) is evaluated at  $8 \times 10^{-6} \, \Omega \cdot m$ .

The theoretical estimation for the sheet resistance ( $R_{sh}$ ) of the complete stack, using the parallel model, is approximately  $28 \, \Omega/\square$ , yielding an effective resistivity of  $2.6 \times 10^{-6} \, \Omega \cdot m$ . Similarly, for the 300-800 nm range, the optimized thicknesses are 45 nm for the top IZO layer and 60 nm for the bottom IZO layer. sheet resistance is  $20.2 \, \Omega/\square$ .

*Note:* Literature indicates that the actual measured resistivity of an IZO/Ag/IZO stack is often lower than the theoretical value calculated using the parallel mixture rule [1]. This reduction is attributed to the formation of an Ohmic contact at the interfaces, given that Ag has a work function of 4.2–4.4 eV, which aligns favorably with the IZO work function of 4.4–5.1 eV [1]. Experimental measurements of stacks without any adhesion treatment or wetting layers have shown a relative resistance reduction of approximately 24% from the initial estimated value [1].

### 2.3.3 Figure of Merit (FoM)

The Figure of Merit (FoM) is a critical metric used to evaluate and select materials for transparent conducting oxides, as it simultaneously accounts for both optical transparency and electrical conductivity.

Haacke’s Figure of Merit is defined as:

$$\text{FoM} = \frac{T^{10}}{R_{sh}} \quad (2.10)$$

Here,  $T$  is the transmittance (expressed as a fractional value). While some studies use the specific transmittance at 550 nm (where solar intensity peaks), we utilized the weighted average transmittance across the entire targeted spectrum alongside the sheet resistance calculated above. For comparison, a standard 140 nm IZO layer was assigned a sheet resistance of  $32 \, \Omega/\square$ .

Table 2.3: Haacke’s Figure of Merit over the 300–800 nm range.

| <b>Sample</b> | <b>Haacke FoM (<math>\Omega^{-1}</math>)</b> |
|---------------|----------------------------------------------|
| IZO (140 nm)  | $2.85 \times 10^{-3}$                        |
| IZO/Ag/IZO    | $5.70 \times 10^{-3}$                        |

Table 2.4: Haacke’s Figure of Merit over the 300–1300 nm range.

| <b>Sample</b> | <b>Haacke FoM (<math>\Omega^{-1}</math>)</b> |
|---------------|----------------------------------------------|
| IZO (140 nm)  | $1.04 \times 10^{-3}$                        |
| IZO/Ag/IZO    | $4.09 \times 10^{-3}$                        |

Through analysis of these three parameters— $J_{sc}$ , sheet resistance, and FoM—the IZO/Ag/IZO multilayer structure demonstrates significant optical and electrical improvements; we can conclude that it is an efficient advanced transparent conducting oxide architecture.

## Chapter 3

# Deposition Method Analysis

In the NCPRE lab, multiple deposition methods are utilized, and most of the silver deposition is performed via evaporation. However, a review of the literature indicates that sputtering is widely favored for such depositions. This naturally raises the question of which technique is optimal for silver nanolayers. A fundamental consideration of cost arises whenever working with noble metals such as silver. It is necessary to evaluate how much the cost of the multilayer TCO will increase compared to a traditional TCO, and how cost-efficient this structure will be relative to the percentage gain in performance.

Although a comprehensive cost analysis must consider the time required for the multistructure to form (due to vacuum system operations), this analysis is limited to material costs only.

### 3.1 Motivation

1. **Analysis for the best method for silver deposition:** Targeting less cost and more uniformity. Material loss is a major consideration. For example, in the evaporation chamber, pointy structures formed of silver can be observed (see Figure 3.1). This formation shows extra silver material depositing on the chamber structures (the exact number of prior depositions is unknown).
2. **Cost comparison:** Answering the question of how much more an IZO (45 nm)/Ag (8 nm)/IZO (40 nm) structure will cost compared to a standard IZO (140 nm) layer, and comparing this cost with the percent gain.



Figure 3.1: Material loss observed in the evaporation chamber (number of depositions-unknown).

## 3.2 Cost Approximation of Silver Deposition

### 3.2.1 Evaporation Tool

The evaporation flux of evaporated material varies with angle, being maximum at the center. The closer the finite-area substrate is to the boat, the wider the solid angle it subtends, leading to more angular variation in thickness. Hence, the substrate is kept at a significantly larger distance compared to sputtering. However, a larger substrate-to-boat distance causes significant material loss.

The flux emitted is given by the following formula for small-area sources [23]:

$$d = \frac{M \cos \theta \cos \phi}{\pi \rho r^2} \quad (3.1)$$

If the substrate is approximately above the boat, this reduces to [23]:

$$m = \pi \rho R^2 t \quad (3.2)$$

Where:

- $\rho$  = Density of the material.
- $R$  = Distance from the boat to the substrate.
- $t$  = Required thickness of the material.

From the above calculations, for  $R = 30$  cm and  $t = 8$  nm of silver, the required mass of material comes out to be 23.7 mg of silver. But for this specific evaporation tool, the tooling factor is 40%. This means the sensor will receive 40% more material, and then it will show 8 nm on the screen (e.g., for a 20 nm thickness on the sensor, it will show 8 nm on the substrate). The final required mass of silver becomes 60 mg.

After finally accounting for dead volume and boat waste, and multiplying the amount by 3.5 (approximate lab multiplier), we approximate the final amount to be around 200 mg per deposition of silver.

### 3.2.2 Sputtering Tool

Unlike the evaporation tool, sputtering utilizes plasma particles, which decrease their mean free path. Where the mean free path was in meters for the evaporation tool (which allowed us to use geometry to formulate the mass reaching the substrate), modern magnetron sputtering tools confine particles in a small region near the target (silver in our case). This requires the substrate to be placed near the target material, so the loss of material is relatively very low.

In our case, most sputtering was done at around an 8 cm distance. Because of these fundamental differences, we cannot use basic geometry for the calculation. Rather, the transport efficiency of the material in the plasma is used:

$$\eta \cdot m_{\text{target}} = m_{\text{film}} \quad (3.3)$$

$$m_{\text{target}} = \frac{A \times t \times \rho}{\eta} \quad (3.4)$$

The variables are defined as:

- $m_{\text{target}}$  = The total mass of silver knocked off the target (in grams).
- $A$  = The surface area of your substrate holder/platen (in  $\text{cm}^2$ ).
- $t$  = Target film thickness (in cm).
- $\rho$  = Density of the material (for Ag,  $10.49 \text{ g/cm}^3$ ).

- $\eta$  = Material transfer efficiency (a percentage expressed as a decimal).

This is calculated using the material transfer efficiency model, as transport in magnetron sputtering is heavily limited by gas scattering [24]. The value of  $\eta$  varies with the pressure inside the chamber; here, it is chosen at 3 mbar. For an 8 nm silver film thickness, an area of the substrate of 7 cm<sup>2</sup>, and a transfer efficiency of 0.5 [24], the mass knocked off the target comes out to be 8.2 mg per 8 nm film deposition.

This is approximately 24 times less than the evaporation mass required per deposition.

### 3.3 Conclusion on Deposition Methods

From the mass required for the same thickness of material, we can conclude the cost of material required for the same thickness in both methods. But it becomes quite complex. Such a calculation requires field knowledge which is insufficient right now. For example, lab-grade silver prices are different from market silver prices, and the target required for sputtering is quite costly compared to the chunks of silver required for evaporation. But for continuous, frequent deposition, sputtering appears cheaper.

One can use the same sputtering calculation for silver and perform an IZO/Ag/IZO and standard IZO cost comparison using the mass calculations above. Also, mass calculations can be used to understand material loss, which becomes essential if frequent silver deposition is required.

In the ultrathin regime, uniformity effects of PVD techniques become more important. And the higher kinetic energy of the sputtering process might give us a smoother film because of surface diffusion [25]. This could be further explored.

## Chapter 4

# Conclusion and Future Outlook

### 4.1 Identified Research Gaps

While simulating for nanothin silver under a 15 nm region, I faced a scarcity of  $n$  and  $k$  data. Although there are a lot of studies regarding nanothin silver layers, most of the papers that work within a 15 nm thickness regime do not provide the corresponding graphs or  $n$  and  $k$  data files. Studying such optical constants and material behavior under the 15 nm regime could be highly useful.

Additionally, for a uniformity comparison between evaporation and sputtering, I could not find a solid study for an ultrathin silver layer. Therefore, investigating the effects of deposition techniques on the morphology of noble metal nanolayers, mostly under a 15 nm thickness, could turn out to be very useful for future research.

### 4.2 Conclusion

In this thesis work, the IZO/Ag/IZO multilayer structure was investigated for application as a transparent conducting oxide layer. In this thesis, a study of optical design and simulation for such multilayer TCO structures was done. By leveraging admittance matching and compensating for reduced charge carriers using a silver nanolayer, we successfully identified optimized thickness configurations for both single-junction and tandem solar cell applications.

For the visible wavelength range (300–800 nm), the optimized structure of IZO (45 nm)/Ag (8 nm)/IZO (60 nm) demonstrated an 8.55% improvement in the theoretical short-circuit current ( $J_{sc}$ ) cap compared to a standard 140 nm IZO layer. For the broader visible and near-infrared range (300–1300 nm), the IZO (45 nm)/Ag (8 nm)/IZO (40 nm) configuration yielded a 13.06% increase in transmittance, corresponding to a 6.17% improvement in the  $J_{sc}$  cap. Additionally, Haacke's Figure of Merit (FoM) for the optimized stacks showed a two-to-fourfold enhancement over traditional structures, confirming the superior electro-optical balance of the multilayer approach.

Finally, a comparative analysis of deposition methods revealed that even though thermal evaporation is commonly utilized in laboratory settings, magnetron sputtering offers significantly higher material transfer efficiency for ultrathin silver layers. Calculations showed that sputtering requires approximately 24 times less target mass per deposition due to minimized angular loss and directional plasma confinement. This makes sputtering a highly cost-effective and scalable choice for fabricating advanced TCO structures. Future work will focus on the physical fabrication, electrical testing, and morphological characterization of these optimized stacks to further validate these theoretical simulations.

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